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Where presidential campaigns go, and what travels online: a pre-registered search of US county election returns, 2016–2024

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What this report adds

What is known. Presidential candidates concentrate their appearances in the populous parts of competitive states, and the persuasive effects of campaign contact in general elections are small or short-lived. US counties linked by Facebook friendships vote more alike.**What this report adds.** Across 9,312 county-elections in 2016, 2020 and 2024, one public event moved its county's two-party vote by less than half a point. Within the states each ticket contested, two determinants of where events were held passed every pre-registered causal criterion: the county's population, and the events it had hosted at the previous election. Seventeen other hypotheses did not. A county's swing tracked the swing of the distant counties it is tied to on Facebook ($r = 0.72$), robustly and without a pre-existing drift, but both swings occur in the same election.**Implication.** The advice to campaign where people live describes what campaigns already do. These data give no vote-based reason to prefer big cities over small states. Whether online ties carry political change, or connected places respond alike to the same campaign, needs data with timing.

Abstract

Objectives. "Contagion theory" holds that national candidates can skip small places, because the internet carries their message there anyway, and should campaign in big cities instead. We tested the premises it rests on, then searched the same data for what determines where campaigns hold events and how county votes move.**Methods.** County presidential returns for 2016, 2020 and 2024 from the MIT Election Data and Science Lab (3,104 counties; 9,312 county-elections) were linked to 863 public events held by the major-party nominees and running mates (FairVote), to Facebook friendship ties between counties (Meta Social Connectedness Index), and to census, metropolitan, airport, unemployment and amenity measures. Every comparison was made within a state's election, with design-based standard errors clustered by state or, for event counts, by county. An automated, version-controlled pipeline tested pre-registered hypotheses in eight phases under Benjamini–Hochberg false-discovery control; associations that cleared a gate fixed in advance were assessed against pre-registered causal criteria.**Results.** An event by the Democratic nominee changed the county's Democratic share of the two-party vote by +0.07 points (95% CI –0.20 to +0.34), and a Republican event by –0.20 points (–0.41 to +0.01); effects of half a point were excluded. Within contested states, each log unit of population added 0.158 events per election (95% CI 0.112 to 0.204), with similar estimates in each election, and each event a county had hosted at the previous election added 0.503 events (0.370 to 0.636) beyond population, partisan lean, closeness and the previous swing. Seventeen further hypotheses failed. A county's swing tracked the swing of its out-of-state online friends ($r = 0.72$; adjusted 3.64 points per point, 95% CI 3.37 to 3.91) and passed a placebo test that physical neighbours failed.**Conclusions.** Campaigns go where people live and return to where they have been, while a visit barely moves its own county's vote. Votes moved along online ties, but these data cannot show that the ties carried the change.

1. Introduction

The claim examined here was stated as follows: candidates running for national office should not bother visiting small states, because with the modern internet information reaches smaller places quickly without in-person contact; they should campaign in large cities instead, where their message spreads further and faster. We refer to it as contagion theory. It is a claim about strategy, but it rests on premises that data can test: that an in-person event moves votes where it is held; that its effect travels to places linked online; that visits matter less where more people are online; that events in large places reach further than events in small ones; and that events in small states add nothing.

Earlier work bears on each premise. Presidential appearances from 1972 to 2000 concentrated in large and dense populations [1]. A field experiment that randomised a governor's schedule found more, and more favourable, local coverage where he appeared [2]; survey evidence finds that rallies shift support for days rather than weeks [3, 4]; and field experiments put the persuasive effect of campaign contact in general elections close to zero [5]. Facebook friendship links between US counties, measured by the Social Connectedness Index [6], are associated with more similar voting [7], although physical proximity to partisans has been found to predict county results better than online ties [8].

This report tests the premises on every US county outside Alaska and Connecticut in three presidential elections. Phase 1 was a confirmatory test. No open, licensed data on in-person campaign visits could be found for any other country, so the later phases searched the same county panel for new correlates and causes of where campaigns go and how votes move, and a second question asked what makes counties grow. The search was given a target of three associations that pass every causal criterion, at least one of them related to contagion, and was ended after the 19 hypotheses of its final stage had produced two. Every hypothesis was declared before it was tested, and every result is reported.

2. Methods

2.1 Data

Election returns. Official precinct returns from the MIT Election Data and Science Lab (MEDSL) were aggregated to counties for 2020 and 2024, and MEDSL's county returns were used for earlier elections. Where precinct sums did not reproduce official totals, county results from an independent compilation were used. The 2024 county totals reproduced MEDSL's official state totals within 0.5% in 46 states, and the 2020 totals reproduced the certified national totals within 0.03%. Alaska, which has no counties, and Connecticut, whose 2024 results are reported by planning region, were excluded, leaving 3,104 counties in 48 states and the District of Columbia. Counts that MEDSL redacts were left missing and never inferred.

Campaign events. FairVote's trackers of public general-election events by the presidential and vice-presidential nominees gave 397, 204 and 262 in-person events in 2016, 2020 and 2024; 862 of the 863 were placed in their county from the place recorded. Fundraisers and private meetings are not tracked, and FairVote's 2012 tracker is not available as data.

Online ties. Meta's Social Connectedness Index gives the relative probability that Facebook users in two counties are friends, for every pair of counties [6]. A county's online friends' swing is the friendship-weighted average swing of counties more than 150 km away, outside its metropolitan area and in another state, so that neither physical spillover nor the county's own state campaign enters. Its physical neighbours' swing weights counties within 150 km by $\exp(-\text{distance}/50 \text{ km})$.

Places. American Community Survey 5-year estimates for 2019–2023 (broadband subscription, education, race and Hispanic origin, household income, age, college enrolment, industry); Census Bureau population estimates and land areas; metropolitan areas from the Office of Management and Budget's 2023 delineation; airports with scheduled airline service (OurAirports); county unemployment rates from the Economic Research Service (ERS) of the US Department of Agriculture, based on Bureau of Labor Statistics data; and the ERS natural amenities scale [9].

2.2 Outcomes and comparisons

Four outcomes were studied: the swing, defined as the change in the Democratic share of the two-party vote since the previous election, in percentage points; the change in votes cast, in percent; the number of public events held in the county by one ticket or by either ticket; and, for the growth question, the percentage change in population between election years (2012–16, 2016–20, 2020–24).

Every estimate compares counties within the same state's election: state-by-election fixed effects were absorbed, which holds fixed battleground status, statewide advertising and down-ballot races. Phase 1 also absorbed county fixed effects. Vote outcomes were weighted by the previous election's votes and growth was unweighted; both used standard errors clustered by state (49 clusters). Studies of events counted only the states that the ticket, or either ticket, visited in that election, through a 0/1 weight that removes unvisited states from both the comparison and the degrees of freedom, and

were clustered by county (2,122 counties in visited states). County clusters were declared after the pipeline's audit found no correlation of events between counties of the same state once state-elections were absorbed (intraclass correlation 0.000); with state clusters, too few states contributed for stable tests.

2.3 The search

The analysis was run by a version-controlled pipeline in four stages. An audit fixes each study's gate from its sample size and design alone, before any exposure is related to an outcome. A cohort stage assembles the rows. A correlation stage tests candidates four at a time with design-based statistics and computes Benjamini–Hochberg q-values [10] over every test ever run in the study's family. A causal stage takes any association that clears the gate (a correlation of at least 0.10 in absolute value within state-elections, with $q \leq 0.10$), assesses it against the causal criteria, and explains it with further in-data analyses and published evidence. All contagion hypotheses from phase 2 onwards share one false-discovery family; the growth question has its own. Every gate was confirmed exactly as the audit proposed it.

2.4 Causal criteria

An association was accepted as a causal case only if it met every criterion declared for its study:

1. the exposure is a condition that a campaign or a county could have; demographic composition was declared a marker of who lives in a county, which can be described but not named a cause;
2. the exposure was settled before the outcome;
3. the association survives adjustment for the confounders declared in advance;
4. it is robust to unmeasured confounding: a confounder as strong as the strongest measured one would not remove it [11]. The robustness value reported is the share of residual variance that a confounder would need to explain in both exposure and outcome to reduce the estimate to zero;
5. it has the same sign in every election, from separate fits per election; and
6. where declared (the network studies), it does not predict the county's change at the previous election, which it could not have caused. A placebo estimate that is significant in the same direction, or at least half the main estimate, fails.

2.5 Pre-registration and deviations

Eleven pre-registrations were committed to the study repository before their studies ran (Table 1). Three deviations occurred, each before the affected results were computed. Phase 3's state-clustered event studies stopped at the pipeline's pre-specified requirement of at least 30 contributing clusters, before any association was computed, and were re-declared with county clusters (phase 3b). For phase 4, the benchmark confounder for population was changed from density to land area in the pre-registration: density is population divided by area, so in phase 3 the benchmark could not be passed. Phase 2 was restarted twice to fix two software faults, after which its final ledger matched the first run test for test. No rule of the causal assessment was changed after a result, and near misses were not re-specified.

Table 1. The phases of the study, each declared and committed before it ran.

Phase	Question	Commit	Result
1	Does a visit move its county's vote and turnout, and does the effect travel online?	efb1461	Under half a point per event; online spillover inconclusive
2	Which county traits moved with the swing and with turnout?	e67af7d	No cause; broadband and distant ties mostly composition
3, 3b	Do votes move along online ties? Which traits drew events?	d4e410f, 434572a	Online co-movement, timing unresolved; population drew events, benchmark saturated
4	What draws events, with land area in place of density?	80b893a	Two causal cases: population, and return visits
5	Return visits re-tested; closeness, swing size, metropolitan core	c942711	Return visits held; three rejected
6	Campus and factory audiences	9121ac9	Both rejected
7	Economic voting, momentum, contesting the other side's ground	bf0c899	All rejected
8	Turnout and competition; growth and the swing	3e69189	All rejected
Growth 1	Does distance to big cities and airports slow county growth?	6cfc342	Both rejected; big-city distance a near miss
Growth 2	Do natural amenities draw growth?	b658938	Below the gate

2.6 Data protection

All data are public aggregates about counties. No person-level data were used, and no figure in this report concerns an identifiable person.

3. Results

3.1 A visit's effect on its own county

Comparing the same county over time within each state's election, an event by the Democratic nominee changed the Democratic share of the two-party vote by +0.07 points (95% CI −0.20 to +0.34) and a Republican event by −0.20 points (−0.41 to +0.01). For both, equivalence tests excluded the pre-registered smallest effect of interest, half a point per event (Figure 1). Compared within state-elections only, without county effects, the Democratic estimate was +0.20 points (0.04 to 0.36), still small. The turnout effect was imprecise (+0.67%, −0.48 to +1.83). Exposure to events through online ties, the moderation of visit effects by broadband adoption and digital advertising, and the reach of events in large compared with small counties were all inconclusive (Table 2). In the smallest states, Republican events had no meaningful effect; the estimate for Democratic events there was inconclusive.

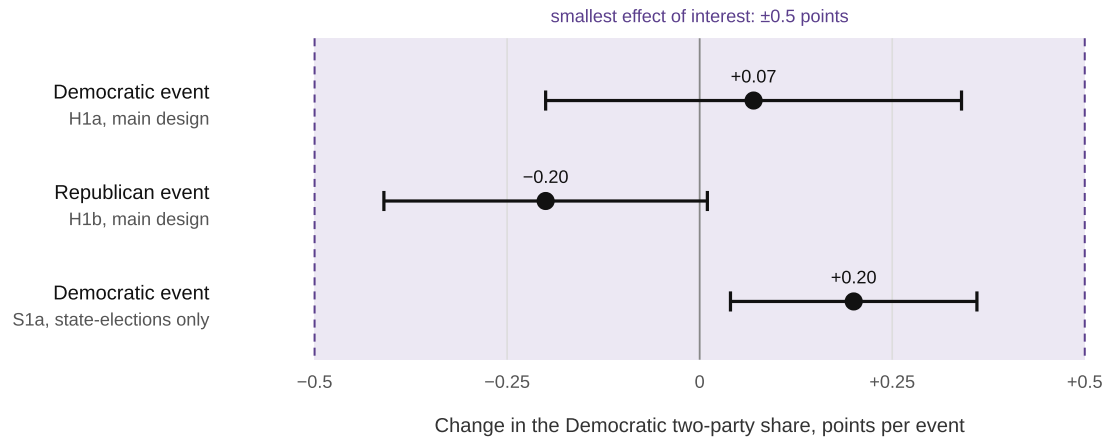


Figure 1. Change in the county's Democratic share of the two-party vote per public event, with 95% confidence intervals. The shaded band is the pre-registered smallest effect of interest (± 0.5 points); equivalence was tested with 90% intervals. H1a and H1b absorb county and state-election effects; S1a absorbs state-election effects only.

Table 2. Phase 1: the premises of contagion theory, pre-registered predictions and verdicts.

Premise	Prediction	Estimate (95% CI)	Verdict
H1 A visit moves its own county	H1a Democratic event raises the Democratic share	+0.07 points (-0.20 to +0.34)	Contradicted: < 0.5 points shown
	H1b Republican event lowers the Democratic share	-0.20 (-0.41 to +0.01)	Contradicted: < 0.5 points shown
	H1c An event raises turnout	+0.67% (-0.48 to +1.83)	Inconclusive
H2 The effect travels online	H2a Ties to Democratic events, Democratic share	-0.34 (-3.0 to +2.3) per unit	Inconclusive
	H2b Ties to Republican events, Democratic share	-3.0 (-7.1 to +1.1)	Inconclusive
	H2c Ties to events, turnout	-7.3 (-23.2 to +8.7)	Inconclusive
H3 Visits matter less online	H3a Event $\times$ broadband adoption, turnout	+0.47 (-0.39 to +1.33) per point	Inconclusive
	H3b Event $\times$ digital ad spending, turnout	+0.47 (-0.19 to +1.12) per \$ per resident	Inconclusive
H4 Big places reach further	Online spillover, large minus small hosts	+27 (-34 to +88)	Inconclusive
H5 Visits add nothing in small states	H5a Turnout	-0.17% (-0.62 to +0.27)	Inconclusive
	H5b Democratic events, Democratic share	+0.52 (-0.13 to +1.17)	Inconclusive
	H5c Republican events, Democratic share	-0.02 (-0.34 to +0.31)	Supported

Verdicts follow the pre-registered rules: Holm adjustment within each outcome, and equivalence by two one-sided tests against the smallest effect of interest. "Contradicted" for H1a and H1b means an effect of half a point per event or more is excluded. Online exposure is the share of a county's distant friendship links that lead to event counties; typical exposure is about 0.3 units.

3.2 Population and where campaigns go

Events were concentrated. Counties of 500,000 people or more hosted 46–49% of all events in each election; Pennsylvania hosted 160 events over the three elections, North Carolina 116, Florida 102, Michigan 88 and Wisconsin 68; and 24, 17 and 19 states saw any event in 2016, 2020 and 2024.

Within the states a ticket visited, population was correlated with events ($r = 0.35$). With land area, income, education, race and ethnicity, age and the previous Democratic share held fixed, each log unit of population added 0.158 events per election (95% CI 0.112 to 0.204; $p = 2.3 \times 10^{-11}$; $n = 9,306$), so a county ten times as populous as another in the same state hosted about 0.36 more events per election. The estimate was similar in 2016 (0.185; 0.129 to 0.241), 2020 (0.152; 0.099 to 0.205) and 2024 (0.130; 0.073 to 0.187) (Figure 2a), and met every causal criterion (Table 3). Its robustness value was 0.165: an unmeasured confounder as strong as the previous Democratic share, the strongest measured one, would leave 0.111 ($t = 4.55$). Measured confounders accounted for 26% of the crude association. Distance to an airport and the size of the media market, which follow from population, were tested separately and were explained by it (Table 6).

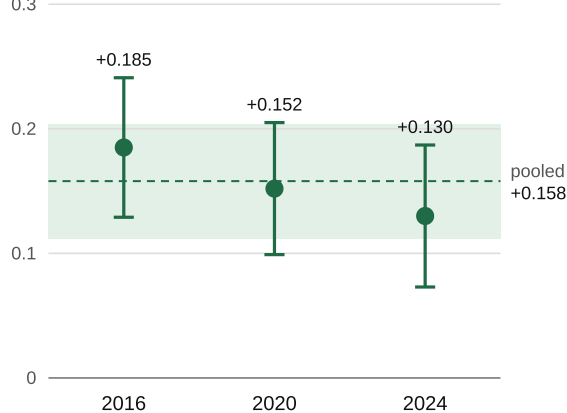
This association had been seen in phase 3, where it failed only the robustness criterion, because its benchmark confounder there, density, is population divided by area. The pass in phase 4, with land area as the benchmark, is therefore a re-test of a seen association under a design changed after that result, and is weaker evidence than a fresh finding would be.

Table 3. Population and campaign events: the causal criteria.

Criterion	Result
Exposure	Population (natural log): the audience a single stop can reach, in person and through local news
Temporal order	Settled before a campaign chooses where to go
Adjusted for confounders	0.158 events per log unit (95% CI 0.112 to 0.204; df 2,121; n 9,306), holding land area, income, education, race and ethnicity, age and the previous Democratic share fixed
Unmeasured confounding	Robustness value 0.165 (0.120 at $\alpha = 0.05$); a confounder as strong as the previous Democratic share would leave 0.111 ($t = 4.55$ against 1.96)
Consistency	2016: 0.185 (0.129 to 0.241); 2020: 0.152 (0.099 to 0.205); 2024: 0.130 (0.073 to 0.187)

(a) Population

events per log unit of population



(b) Return visits

events per event hosted at the previous election

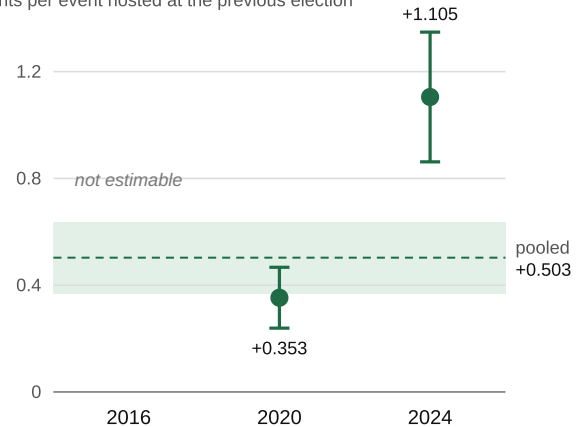


Figure 2. The two causal cases, election by election, with 95% confidence intervals. (a) Events per log unit of population. (b) Events per event hosted at the previous election (phase 5 model); 2016 cannot be estimated because FairVote's 2012 tracker is not available as data. Dashed lines and shaded bands show the pooled estimate and its interval.

3.3 Return visits

Phase 4 tested whether campaigns return to places they visited at the previous election, beyond everything that made those places attractive the first time. Each event a county had hosted at the previous election added 0.493 events (95% CI 0.361 to 0.625; $p = 3.7 \times 10^{-13}$), with population, land area, airport distance, media-market size, income, education, race and ethnicity, age and the previous Democratic share held fixed. Phase 5 re-tested it with its most obvious rival explanations added: the closeness of the previous result, the size of the previous swing, and whether the county is its metropolitan area's central county. The estimate held (0.503; 0.370 to 0.636; $p = 1.9 \times 10^{-13}$), in 2020 (0.353; 0.239 to 0.467) and in 2024 (1.105; 0.862 to 1.348) (Figure 2b, Table 4). Its robustness value was 0.464: to reduce it to non-

significance, an unmeasured factor would have to explain 37% of the remaining variation in both earlier and current events, whereas population, the strongest measured factor, explains 12% and 5%. Counties with no event, one event, and two or more events at the previous election held -0.10 , $+0.35$ and $+2.07$ events relative to their state's average.

Table 4. Return visits: the phase 4 test and the phase 5 re-test.

Criterion	Phase 4	Phase 5 re-test
Adjusted effect	0.493 events per earlier event (0.361 to 0.625)	0.503 (0.370 to 0.636)
Confounders	Population, land area, airport distance, media-market size, income, education, race and ethnicity, age, previous Democratic share	The same, plus closeness and size of the previous swing, and metropolitan core
Unmeasured confounding	Robustness value 0.459; a confounder as strong as population would leave 0.425	Robustness value 0.464; would leave 0.442
Consistency	2020: 0.351 (0.238 to 0.464); 2024: 1.102 (0.858 to 1.346)	2020: 0.353 (0.239 to 0.467); 2024: 1.105 (0.862 to 1.348)

Both models: within the states either ticket visited, 2020 and 2024 ($n = 6,204$), standard errors clustered by county ($df\ 2,121$).

3.4 Votes and online ties

Phase 3 related a county's swing to the swings of the places it is tied to (Table 5). Within state-elections, the swing correlated with the online friends' swing at $r = 0.72$, and with the physical neighbours' swing at $r = 0.47$. With the county's own traits and the other channel held fixed, one point of online friends' swing went with 3.64 points of the county's own (95% CI 3.37 to 3.91; $p = 1.4 \times 10^{-30}$), about 1.7 points per standard deviation. The coefficient is large because the friends' average varies little within a state (standard deviation 0.47 points, against 3.0 for a county's own swing). The association held in each election, was robust to unmeasured confounding (robustness value 0.59), and ran in the opposite direction for the county's previous swing (placebo -0.85 ; -1.61 to -0.10). Neighbours' swing also survived adjustment (0.36 per point; 0.25 to 0.47), but it predicted the county's previous swing (0.24; 0.03 to 0.44), two-thirds of the main estimate, a sign of a drift that predates the election.

Both were rejected as causes because the two swings occur in the same election. The version with time order built in, the online friends' swing at the previous election against the county's current swing, correlated at $r = 0.18$ and was explained by the county's own earlier swing (0.35 points per point; -0.43 to 1.13). Turnout did not travel along online ties ($r = 0.11$; adjusted 0.77, -0.67 to 2.21).

Table 5. A county's swing and the swings of the places it is tied to (phase 3).

	Online friends' swing	Physical neighbours' swing
Correlation within state-elections	$r = 0.72$	$r = 0.47$
Adjusted, per point (95% CI)	3.64 (3.37 to 3.91); 1.7 points per SD	0.36 (0.25 to 0.47); 0.5 points per SD
2016 / 2020 / 2024	1.29 (0.87 to 1.71) / 2.41 (0.80 to 4.03) / 0.98 (0.36 to 1.59)	0.28 (0.15 to 0.41) / 0.27 (0.16 to 0.38) / 0.33 (0.06 to 0.61)
Placebo: the county's previous swing	-0.85 (-1.61 to -0.10): opposite sign, passes	0.24 (0.03 to 0.44): two-thirds of the main estimate, fails
Unmeasured confounding	Robust (robustness value 0.59)	Not robust
Verdict	Not a demonstrated cause: same election	Not a demonstrated cause: same election, shared drift

Adjusted for the county's education, race and ethnicity, age, income, density, size, growth, broadband and distant ties, and for the other channel. Standard errors clustered by state ($df\ 48$); $n = 9,234$.

3.5 The other seventeen hypotheses

Seventeen further hypotheses were tested in phases 4 to 8 and in the growth question (Figure 3, Table 6). Ten cleared their gates (twelve counting each ticket's study separately), and each failed a causal criterion; the other seven did not reach their gates. Distance to an airport, the size of the media market and the metropolitan core were explained by population, and contesting the other ticket's ground was explained by each ticket's own earlier events.

Two failures were informative. Population growth was associated with the Democratic swing after adjustment (0.174 points per percentage point of growth; 0.126 to 0.222), but it changed sign between elections (2016: -0.029 , -0.116 to 0.058 ; 2020: 0.067 , 0.027 to 0.107 ; 2024: 0.021 , -0.002 to 0.044); it re-tested an association seen in phase 2. In the growth question, distance to the core of a metropolitan area of a million or more went with slower growth in every period, and more strongly over time (2012–16: -0.08 , -0.26 to 0.11 ; 2016–20: -0.20 , -0.40 to -0.003 ; 2020–24: -0.35 , -0.71 to 0.003), the largest in the pandemic years. With size, land area, income, education, make-up and age held fixed, it missed significance (-0.213 points of growth per log kilometre; -0.435 to 0.009 ; $p = 0.059$) and was not robust, so it was rejected and was not re-specified.

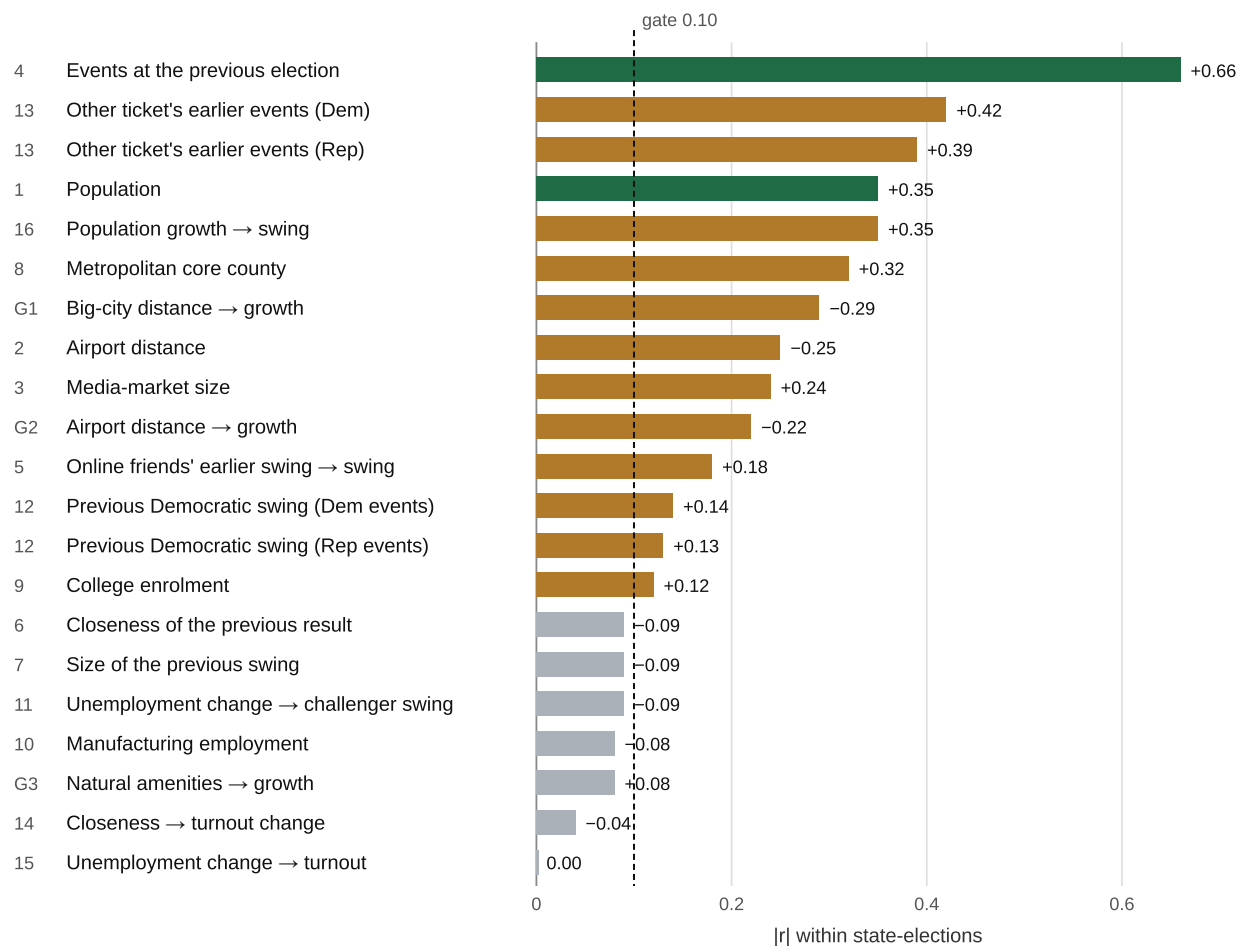


Figure 3. Screening correlation of each hypothesis in the search for three causes, within state-elections (state-periods for growth), as an absolute value with the sign printed, against the gate of 0.10. The outcome is the county's public events unless named. Green: causal case; ochre: rejected after clearing the gate; grey: below the gate.

Table 6. Every hypothesis in the search for three causes, with its screening correlation, adjusted effect and verdict.

#	Exposure	Outcome	r	Adjusted effect (95% CI)	Verdict
Phase 4 · events within visited states, clustered by county					
1	Population (log)	events	+0.35	+0.158 (0.112 to 0.204)	Causal case
2	Distance to an airport with scheduled service (log km)	events	-0.25	+0.004 (-0.064 to 0.072)	Confounded by population
3	Media-market population (log)	events	+0.24	+0.005 (-0.009 to 0.019)	Confounded by population
4	Events at the previous election	events	+0.66	+0.493 (0.361 to 0.625)	Causal case
5	Online friends' swing at the previous election	swing	+0.18	+0.349 (-0.431 to 1.129)	Confounded by own earlier swing

#	Exposure	Outcome	r	Adjusted effect (95% CI)	Verdict
Phase 5					
6	Closeness of the previous result	events	-0.09	not estimated	Below the gate
7	Size of the previous swing	events	-0.09	not estimated	Below the gate
8	Central county of its metropolitan area	events	+0.32	-0.048 (-0.189 to 0.093)	Confounded by population
4'	Hypothesis 4, with 6–8 added as confounders	events	+0.66	+0.503 (0.370 to 0.636)	Causal case holds
Phase 6					
9	Share of residents enrolled in college (%)	events	+0.12	-0.006 (-0.015 to 0.003)	Confounded
10	Share of workers in manufacturing (%)	events	-0.08	not estimated	Below the gate
Phase 7					
11	Change in unemployment over the incumbent's term	swing to challenger	-0.09	not estimated	Below the gate
12	Previous swing toward the Democrats	Dem / Rep events	+0.14 / +0.13	-0.028 (-0.038 to -0.018) / -0.015 (-0.023 to -0.007)	Confounded: sign reversed
13	Other ticket's events at the previous election	Dem / Rep events	+0.42 / +0.39	+0.055 (-0.071 to 0.181) / +0.059 (-0.081 to 0.199)	Confounded by own earlier events
Phase 8					
14	Closeness of the previous result	turnout change	-0.04	not estimated	Below the gate
15	Change in unemployment	turnout change	+0.00	not estimated	Below the gate
16	Population growth (%), re-test	swing	+0.35	+0.174 (0.126 to 0.222)	Not consistent across elections
Growth question · within states, unweighted, clustered by state					
G1	Distance to the core of a metro area of 1 million or more (log km)	growth	-0.29	-0.213 (-0.435 to 0.009)	Not significant; not robust
G2	Distance to an airport with scheduled service (log km)	growth	-0.22	+0.080 (-0.126 to 0.286)	Confounded by big-city distance and size
G3	Natural amenities scale	growth	+0.08	not estimated	Below the gate

r: screening correlation within state-elections. Adjusted effects are in outcome units per unit of exposure (events; percentage points of swing or growth). "Not estimated": the association did not clear its gate ($|r| \geq 0.10$), so no adjusted model was fitted. Hypotheses 12 and 13 were tested for each ticket's events in 2020 and 2024.

4. Discussion

4.1 Principal findings

Within the states they contested, both campaigns chose where to appear by population, and returned to where they had appeared before. A single visit moved its own county's vote by less than half a point. A county's vote moved with the votes of distant places linked to it by online friendships, more than with its physical neighbours and without a pre-existing drift, but within the same election.

4.2 Implications for contagion theory

The prescription to campaign in big places is already practice. The theory's premise about information has some support: the co-movement of votes along online ties is what an online spread of political mood would produce. A shared response of connected places to the same national campaign would produce the same pattern, and these data cannot separate the

two. The case for preferring big cities over small states cannot rest on the vote effect of a visit, which is small everywhere. Campaigns are also creatures of habit: a place visited once is likely to be visited again, beyond anything measured about it.

4.3 Comparison with other studies

The concentration of events in populous counties agrees with the analysis of appearances from 1972 to 2000 [1]. In 2008, one campaign avoided growing counties and the other sought them [12]; here, growth added nothing once size was known. Field offices, which co-locate with visits and can stay in place across elections, raised county vote shares by about a point in 2008 [13] and are one candidate explanation for return visits. The small effects of appearances agree with studies of rallies [3, 4] and with field experiments on campaign contact [5], although a randomised schedule did increase local coverage [2]. The co-movement of votes along online ties agrees with evidence that connected counties vote alike [7]; the contrasting finding that physical proximity to partisans predicts county results better than online ties [8] concerns levels of support rather than changes between elections. The penalty of distance from large cities on county growth agrees with findings for 1950–2000 [14].

4.4 Strengths and limitations

The study covers every county of 48 states and the District of Columbia in three elections, uses official returns validated against certified totals, declared every hypothesis before testing it, kept one false-discovery family across the search, and reports every result, including the failures. It has seven limitations:

- The data are ecological: county results cannot say which voters moved, or whether minds changed or the electorate did.
- Visits travel with other campaign investment, such as field offices, local advertising and canvassing, which is not measured by county.
- The event studies treat counties of a state as independent once its election is held fixed, as the audit found; a regional tour could violate this and make those tests somewhat too confident.
- Event counts are mostly zeros and were analysed with linear models.
- FairVote tracks public events only, and its 2012 tracker is not available as data.
- The Social Connectedness Index is a 2026 snapshot and the census measures are 2019–2023 averages, later than some of the elections; they are treated as standing conditions.
- The population finding re-tests an association seen earlier, under a benchmark changed after that result.

4.5 What would settle it

A field experiment that randomises where a national campaign appears [2]; campaign scheduling records showing which places were considered and why; attention data, such as search interest, news mentions or polling by media market week by week, to see whether shifts move along online ties with a lag; county measures of other campaign investment; and more elections.

5. Conclusions

In US presidential elections from 2016 to 2024, campaigns held their events where people live and returned to places they had visited before, and a single visit barely moved its own county's vote. County votes moved with distant places linked by online friendships. Whether those ties carry political change is the question these data leave open.

Declarations

Ethics. All data are public aggregates about counties. No person-level data were used, and no ethics approval was required.

Data availability. All data are public; the sources and their licences are listed below.

Code availability. The analysis code, the eleven pre-registrations, and the audits, ledgers and verdicts of every study are kept in the author's version-controlled study repository; the commit of each pre-registration is given in Table 1.

Funding. None.

Competing interests. None declared.

Use of artificial intelligence. Analyses were executed by a deterministic, version-controlled analysis pipeline. The pipeline was operated, and the pre-registrations, literature searches and this report were drafted, with the assistance of a large language model (Claude, Anthropic). All statistical output was produced by the pipeline.

Data sources

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